

mmWave CSI-based Sensing: a Feasibility Study

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Motivation

Joint Communication and Sensing (JCAS)

- Emerging technique that enables simultaneous communication and environmental sensing
- The Channel State Information (CSI) of the received signal can capture the electromagnetic footprint of the environment

Millimeter Wave (mmWave) Communications

- Support wider bandwidths, enabling high-capacity communication
- Shorter wavelengths enable directional beams and spatial multiplexing for efficient, dense deployments.

High-resolution sensing and high-speed communications

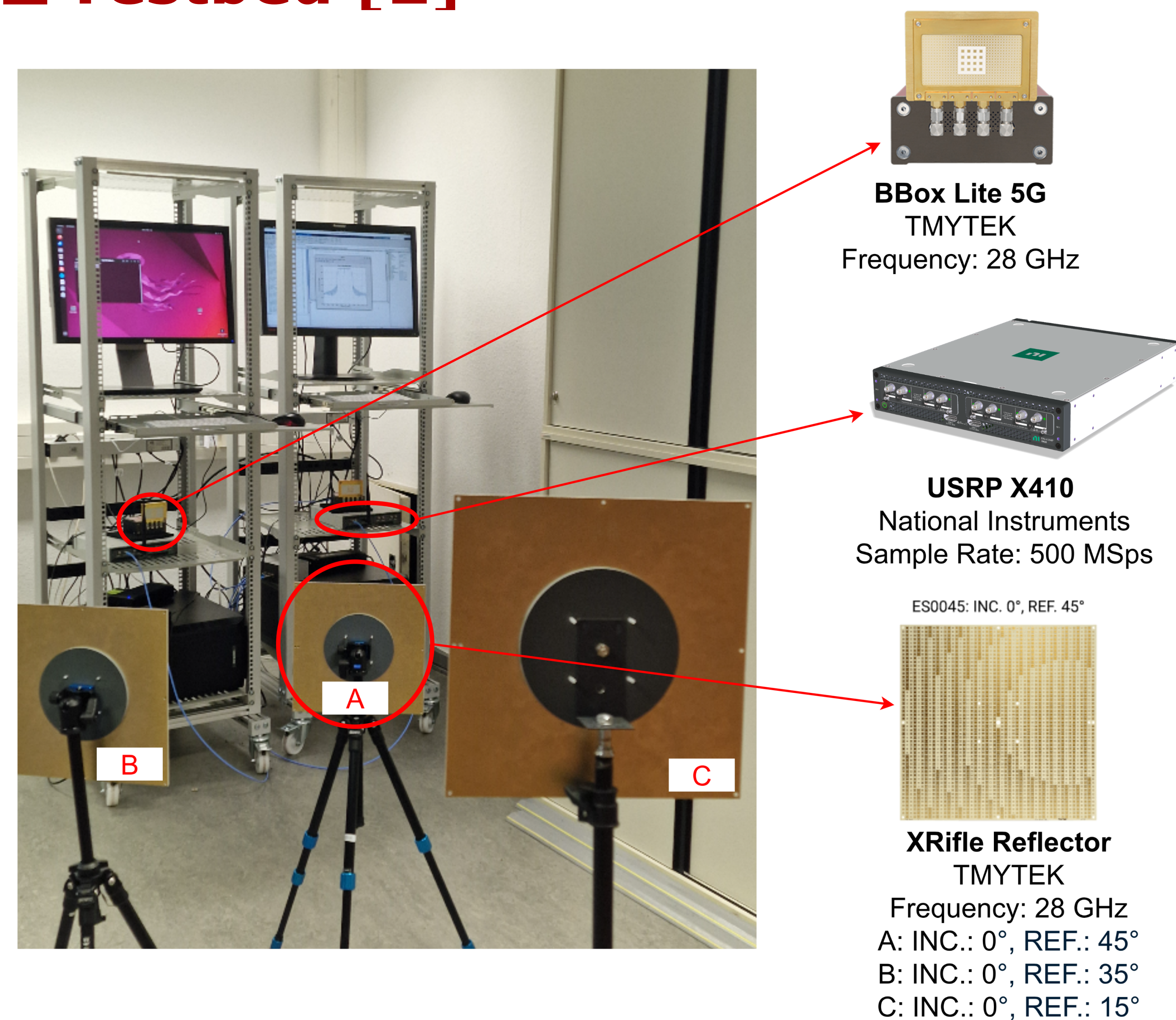
Key Research Question

Can a quantitative analysis of the CSI at 28 GHz be used to distinguish different scenarios of a room (e.g., empty room, the presence of a person standing or walking)?

CSI Processing Algorithm [1]

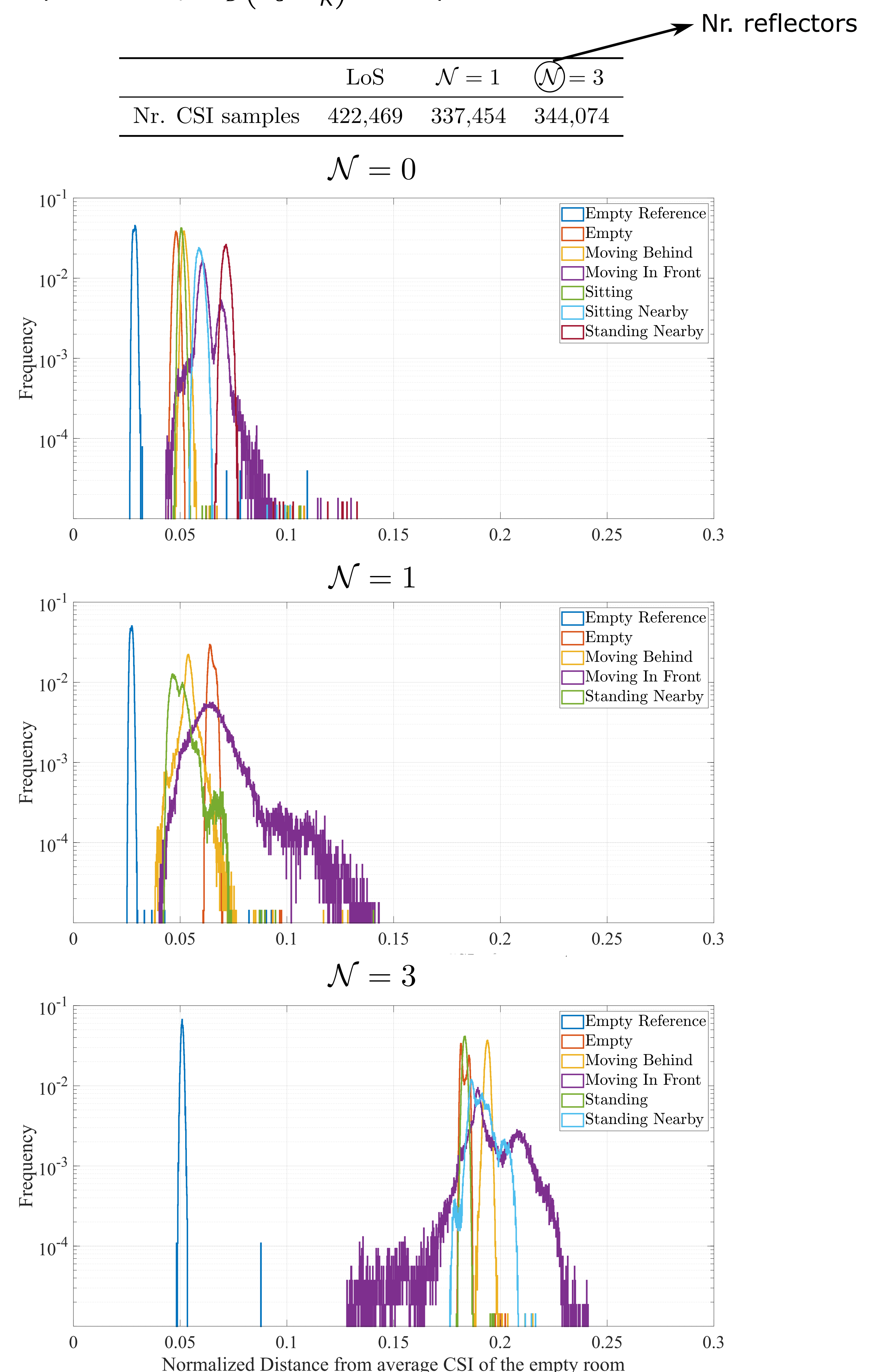
- Extract CSI samples from received packets
 $C(n) = A_C(n)e^{j\angle C(n)}$
- Compute global maximum and minimum and map the amplitude onto the $[0,1]$ interval, then **quantize** the values using 9 bits.
- Compute the **normalized** distances
$$W_D(A_{C_i}, A_{C_j}) = \frac{1}{N_{SC}(2^{q_{amp}} - 1)} \sum_{n \in SC} |A_{C_i}(n) - A_{C_j}(n)|$$
- Estimate the **similarity** between the CSIs of an experiment and the average CSI of one reference experiment
$$W_D(A_{C_i}, A_R^*)$$

Testbed [2]



Results

A reference empty-room experiment A_R is selected for each set of experiments., For every experiment C , $W_D(A_C, A_R^*)$ is computed.



Main Takeaways

- Perfectly static environment (Empty Room):
CSIs are **very similar** to each other
- Person moving among the communication nodes:
Significant modifications to A_C and **greater** W_D .

Future Work

- A more extensive analysis using a larger dataset could yield promising results for CSI-based mmWave sensing, enabling clearer differentiation between a wider range of scenarios.
- This study considers only the amplitude of the CSI; incorporating phase information as well could further enhance accuracy and distinguishability.

References

- [1] E. Tonini, F. Gringoli, R. A. Lo Cigno, and M. Cominelli, "Towards a Quantitative Analysis of CSI for AI/ML Based Sensing", IEEE WCNC, Milan, Italy, March 2025, pp. 1-6.
- [2] A. Zubow, J. Angjo, S. Dimce, and F. Dressler, "Achieving Resilience in mmWave Communications by Using Non-reconfigurable Reflecting Surfaces", IEEE WCNC, Milan, Italy March 2025, pp. 1-6



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